CBCS SCHEME

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BCS306B

ACHARYA INSTITUTE OF TECHNOLOGY BANGALORE MCA Library Date

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Object Oriented Programming with C++

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is meant by constructor? Discuss the types of constructors in C++ with example.	07	L2	CO1
	b.	Demonstrate with example, how friend functions, friend classes and inline functions are useful in C++.	07	L2	CO1
	c.	With a sample program, demonstrate the general format of a C++ program.	06	L3	CO1
		OR			
Q.2	a.	List the various access specifiers supported by OOPs. Illustrate their use.	07	L2	CO1
	b.	What is the significance of static data and member functions in C++? Explain.	07	L2	CO1
	c.	What is Object Oriented Programming? Explain its features.	06	L2	CO1
		Module – 2			
Q.3	a.	Develop a object oriented program to find the smallest and biggest among array elements.	07	L3	CO2
	b.	Discuss pointers to object with example code. Also discuss their advantages.	07	L2	CO2
	c.	Explain different array type in C++ with suitable example code snippet.	06	L3	CO2
		OR			
Q.4	a.	What is "this" pointer? Illustrate the use of "this" pointer in C++.	07	L2	CO2
	b.	Discuss when two or more functions are said to be overloaded. Identify the causes of ambiguity in function overloading.	07	L2	CO2
	c.	What is a dynamic constructor? Explain with an example program.	06	L3	CO2
		Module – 3			
Q.5	a.	List the operators in C++ that can not be overloaded. Develop a C++ program using "Time" class, to overload the '+' and '-' operators.	07	L3	CO3
	b.	Illustrate the role of access-specifiers in different level of inheritances.	07	L2	CO3
	c.	Discuss various types of inheritances with suitable example codes.	06	L3	CO3
		OR			
Q.6	a.	Illustrate the use of constructors and destructors in inheritance in C++.	07	L2	CO3
	b.	Develop a program in C++ to derive a class "Manager" from class "Person" and "Employee". Consider suitable data members and member-functions for the classes.	07	L3	CO3
	c.	Explain "Virtual base Class" with an example.	06	L3	CO2

		Module – 4			
Q.7	**a.**	Design a C++ program demonstrating the use of the Pure Virtual function using base and derived classes. Also explain the code.	**07**	**L3**	**CO4**
	b.	What is polymorphism in C++? Explain its types with example.	**07**	**L3**	**CO4**
	c.	Explain virtual function in C++. Discuss what is early and late binding.	**06**	**L2**	**CO4**
		OR			
Q.8	**a.**	What are generic functions? Demonstrate the use of generic function in swapping two variables of any type.	**07**	**L2**	**CO4**
	b.	What is typename and export keyword? Discuss their usage. Discuss the advantages of using templates in C++.	**07**	**L2**	**CO4**
	c.	Explain the use of a class template. Also explain class template with suitable code snipplet.	**06**	**L3**	**CO4**
		Module – 5			
Q.9	**a.**	Explain the fundamentals of exception handling in C++. Analyze the benefits of exception handling.	**07**	**L2**	**CO5**
	b.	Discuss different standard exceptions in C++.	**07**	**L2**	**CO5**
	c.	Develop a C++ program to demonstrate the usage of try, catch and through to handle exceptions.	**06**	**L3**	**CO5**
		OR			
Q.10	**a.**	What are the different file opening modes in C++? Compare and contrast file opening modes.	**07**	**L2**	**CO5**
	b.	Explain file streams with example.	**06**	**L3**	**CO5**
	c.	Develop a C++ program to create a text file, check file is created or not, if created, write some text in to the file and the read and display the text from the file.	**07**	**L3**	**CO5**

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BCS306B

Third Semester B.E./B.Tech. Degree Examination, June/July 2025
Object Oriented Programming with C++

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define inheritance and explain it with an example.	10	L2	CO1
	b.	Define Object-Oriented Programming. List and explain four main principles of object oriented programming.	10	L2	CO1
		OR			
Q.2	a.	Explain with an example how objects are passed to the function.	10	L2	CO1
	b.	Explain the concept of friend functions in C++. Provide a code example to illustrate how a friend function can access private members of a class.	10	L3	CO1
		Module – 2			
Q.3	a.	Explain with an example the following terms: i) Pointers to derived types ii) Pointers to class members	10	L2	CO2
	b.	Explain the use of arrays of objects in C++ with an example.	10	L3	CO2
		OR			
Q.4	a.	Explain function overloading with an example.	10	L3	CO2
	b.	Explain with an example the following terms : i) Copy constructors ii) Default function arguments	10	L2	CO3
		Module – 3			
Q.5	a.	What is operator overloading? Write a C++ program to illustrating overloading NEW and DELETE operators.	10	L3	CO3
	b.	Define a member operator function in C++ to overload the '+' operator for adding two objects of a class. Explain with code.	10	L2	CO3
		OR			
Q.6	a.	Explain the different types of inheritances used in C++, with an example.	10	L3	CO3
	b.	How does inheritance enable code reusability? Explain with an example.	10	L3	CO3
		Module – 4			
Q.7	a.	What are virtual functions? Describe the rules for declaring virtual function.	10	L2	CO4
	b.	Explain early binding and late binding with example.	10	L3	CO4
		OR			
Q.8	a.	Explain the concept of generic function using templates in C++ with an example.	10	L3	CO4
	b.	What is significance of 'type name' and 'export' in C++ templates? Explain with examples.	10	L2	CO4
		Module – 5			
Q.9	a.	How do you handle exceptions in derived classes? Explain with an example.	10	L2	CO5
	b.	What are different exception handling options in C++? Explain with examples.	10	L3	CO5
		OR			
Q.10	a.	How can you handle errors while reading / writing files in C++? Provide an example.	10	L3	CO5
	b.	How do you detect the End-of-File (EoF) in C++? Discuss with a program.	10	L2	CO5

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CBCS SCHEME - Make-Up Exam

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BCS306B

Third Semester B.E/B.Tech. Degree Examination, June/July 2025
Object Oriented Programming with C++

Time: 3 hrs. Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
1	a.	Define object oriented programming and explain the characteristics of object oriented languages and write general form of C++ program.	10	L1 L2	CO1
	b.	Explain the following with an example : i) Scope resolution operator ii) Friend function	10	L2	CO1
		OR			
2	a.	Describe how objects are passed to functions with an example.	10	L2 L3	CO1
	b.	What are constructors? Explain different types of constructors with an example.	10	L2	CO1
		Module – 2			
3	a.	Develop a C++ program using classes to display student name, roll numbers, marks obtained in two subjects and total score of student.	10	L3	CO2
	b.	Illustrate the use of "this" pointer in C++ with relevant example.	5	L2	CO2
	c.	Explain with example, copy constructor.	5	L2	CO2
		OR			
4	a.	Implement C++ program to sort the elements in ascending and descending order.	10	L3	CO2
	b.	Explain array of pointers to objects and accessing member functions using pointers.	10	L3	CO2
		Module – 3			
5	a.	Discuss the overloading operator using keywords 'new' and 'delete' with an example.	10	L3	CO3
	b.	Develop a C++ program using operator overloading for overloading unary minus operator.	10	L3	CO3
		OR			
6	a.	Develop a C++ program derive a class "Project Lead" from class "Person" and "Employee". Consider suitable data members and member functions for the classes.	10	L3	CO4
	b.	How constructors and destructors get executed for inheritance when there are more than one class and both base and derived classes contain constructors?	10	L3	CO4
		Module – 4			
7	a.	Demonstrate the hierarchical nature of virtual functions with a suitable example program.	10	L2	CO4
	b.	Explain the difference between early and late binding with code example.	10	L3	CO4

		OR			
8	a.	Enumerate the capabilities of Templates.	5	L2	CO5
	b.	Discuss the generic class with an example program.	10	L2 L3	CO5
	c.	Examine the usage of type name and export keywords.	5	L2	CO5
		Module – 5			
9	a.	Develop a function which flows a division by zero exception and catch it in catch block. Write a C++ program to demonstrate usage of try, catch and throw to handle exception.	10	L3	CO5
	b.	Discuss catching all exceptions and rethrowing an exception with examples.	10	L2	CO5
		OR			
10	a.	Explain the following formatted I/o : i) Width ( ) ii) Precision ( ) iii) fill ( ) iv) setf ( ) v) unsetf ( )	10	L2	CO6
	b.	Develop a C++ program to create a text file, check whether the file is created or not. If created, it will write some text into the file and then read the text from the file.	10	L3	CO6

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BCS306B

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Object Oriented Programming with C++

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is object oriented programming? Compare object oriented programming with procedure oriented programming concept.	10	L1 L2	CO1
	b.	Explain the following with an example: i) Scope resolution operator ii) Friend function	10	L2	CO1
		OR			
Q.2	a.	Explain with an example: i) Friend Class ii) Static class members	10	L2	CO1
	b.	What are constructors? Explain different types of constructors with an example.	10	L1 L2	CO1
		Module – 2			
Q.3	a.	Define Class and Object. Develop C++ program to find the largest of three numbers.	08	L3	CO2
	b.	Explain with example, copy constructors.	06	L2	CO2
	c.	Demonstrate with an example of pointers to an objects.	06	L2	CO2
		OR			
Q.4	a.	Implement C++ program to sort the elements in ascending and descending order.	08	L3	CO2
	b.	Explain with example program for Function overloading.	06	L2	CO2
	c.	Demonstrate the concept of this pointer with an example program.	06	L2	CO2
		Module – 3			
Q.5	a.	What is Operator overloading? Explain the overloading of unary operator with an example program.	08	L2	CO3
	b.	What is inheritance? Illustrate the different types of inheritance and syntax of defining derived classes.	08	L2	CO3
	c.	Write short notes on Granting Access.	04	L2	CO3
		OR			
Q.6	a.	Explain the concept of New and delete functions in overloading with an example.	08	L2	CO3
	b.	Demonstrate the concept of inheriting multiple Base classes with example.	08	L2	CO3
	c.	Short notes on Destructors with an example.	04	L2	CO3
		Module – 4			
Q.7	a.	What is polymorphism? How Early and Late Binding in polymorphism with an example program.	10	L2	CO4
	b.	Explain how virtual functions are hierarchical with suitable example program.	10	L2	CO4
		OR			
Q.8	a.	Explain Generic functions with an example program.	10	L2	CO4
	b.	Demonstrate concept of power of templates with an example program.	10	L2	CO4

		Module – 5			
Q.9	a.	Implement a C++ program to create a text file, check file created or not. If created it will write some text into a file and then read the text from the file.	08	L3	CO5
	b.	Develop a function which through a division by zero exception and catch it in catch block, write a C++ program to demonstrate usage of try, catch and through to handle exception.	08	L3	CO5
	c.	Write short notes on formatted I/O functions.	04	L2	CO5
		OR			
Q.10	a.	Implement a C++ program to write and read time in/from binary file using fstream.	08	L3	CO5
	b.	Develop a C++ program that handles array out of bounds exception using C++.	08	L3	CO5
	c.	Write short notes on Derived Class exception.	04	L2	CO5

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BCS306B

Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Object Oriented Programming With C++

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	List and explain the principles of object oriented programming.	6	L2	CO1
	b.	Define Class. Write the general syntax of class. Explain with an example program.	6	L2	CO1
	c.	Describe friend function with an example program.	8	L2	CO2
		OR			
Q.2	a.	What is constructor? Explain parameterized constructor with example.	6	L2	CO2
	b.	Write a short note on : (i) Scope Resolution operator. (ii) Returning objects	6	L2	CO2
	c.	Explain Inline functions with suitable example.	8	L2	CO2
		Module – 2			
Q.3	a.	Explain arrays of objects to pointers with example program.	10	L2	CO2
	b.	Explain : (i) this pointer (ii) Dynamic Allocation operators.	10	L2	CO2
		OR			
Q.4	a.	What is function overloading? Explain function overloading for the following prototype: (i) add(int a, int b) (ii) add (double a, double b)	10	L3	CO3
	b.	Explain copy constructor with suitable example program.	10	L2	CO3
		Module – 3			
Q.5	a.	Explain operator overloading using a friend function.	10	L3	CO3
	b.	Discuss the role of access specifiers in inheritance with example.	10	L2	CO4
		OR			
Q.6	a.	Explain how and where the constructors are used in inheritance with example.	10	L3	CO4
	b.	Define virtual class. Explain how it is used in multiple inheritance with an example program.	10	L2	CO4
		Module – 4			
Q.7	a.	What is virtual function? Demonstrate virtual function with an example program.	10	L3	CO4
	b.	Explain : (i) Early binding V/s Late binding (ii) Pure virtual functions.	10	L2	CO4
		OR			
Q.8	a.	What are generic functions? Explain the generic function that swap 2 variables with which it is called?	10	L3	CO5
	b.	Write a note on : (i) Generic classes (ii) Type name and export.	10	L2	CO5

		Module – 5			
Q.9	a.	What is exception handling? Write a C++ program to throws a division by zero exception and catch it in catch block.	10	L3	CO5
	b.	Explain array out of bounds exception with example program.	10	L3	CO5
		OR			
Q.10	a.	Explain : (i) width ( ) (ii) precision ( ) (iii) fill ( ) (iv) set f( ) (v) unset f( ).	10	L3	CO6
	b.	Write a C++ program to write and read time in/from binary file using fstream.	10	L3	CO6

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CBCS SCHEME

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BCS306B

Third Semester B.E./B.Tech Degree Supplementary Examination, June/July 2024

Object Oriented Programming with C++

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is object oriented programming? How it differs from procedure oriented programming? Explain the structure of C++ program.	10	L1	CO1
	b.	Define constructor. List any 6 characteristics. Explain the concept with program.	10	L2	CO1
		OR			
Q.2	a.	Discuss the importance of class. With suitable example demonstrate the relationship between class and objects.	10	L2	CO1
	b.	Illustrate the process of passing objects of functions with example and explain.	10	L2	CO1
		Module – 2			
Q.3	a.	What is pointer to an object? Demonstrate how to declare and use of pointers to an objects in C++.	10	L2	CO2
	b.	Explain the following with example : i) Function overriding ii) The "this" pointer.	10	L2	CO2
		OR			
Q.4	a.	Write a C++ program to illustrate function overloading with proper output.	10	L3	CO2
	b.	Explain the concept of ambiguity in function overloading. How it can be resolved.	10	L2	CO2
		Module – 3			
Q.5	a.	What is Inheritance? List and explain any 4 types of it.	10	L1	CO3
	b.	With suitable program, explain the concept of operator overloading.	10	L2	CO3
		OR			
Q.6	a.	With proper output, write a C++ program to demonstrate usage of 'new' and 'delete' operators.	10	L3	CO4
	b.	Explain the use of "protected" members in inheritance with program.	10	L2	CO4

		Module – 4			
Q.7	**a.**	What are virtual functions? With program, explain how virtual attribute is inherited.	**10**	**L2**	**CO4**
	b.	Define 'Pure virtual functions'. Differentiate 'Virtual functions' and 'pure virtual functions' with example.	**10**	**L2**	**CO4**
		OR			
Q.8	**a.**	Explain the differences between early binding and late binding.	**10**	**L2**	**CO4**
	b.	Discuss the following : i) Use of templates in C++ ii) Type name and export keyword.	**10**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Explain the fundamentals of exception handling with suitable program.	**10**	**L2**	**CO5**
	b.	How do you handle 'Derived Class Exceptions' explain with example?	**10**	**L2**	**CO5**
		OR			
Q.10	**a.**	Explain the following : i) Formatted IO ii) File streams and string streams.	**10**	**L1**	**CO6**
	b.	Discuss any 4 file handling methods.	**10**	**L2**	**CO6**

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